

Hailey Garcia

jgarc86@lsu.edu | haileyjay.net

Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803

Education

Ph.D. Mathematics, Louisiana State University Fall 2025–present

Advisor: Christin Bibby

B.S. Mathematics, Louisiana State University Fall 2022–Spring 2025

Papers and Preprints

H. J. Garcia. *Classification of strictly resistance nonnegative graphs*. (2026). [arXiv:2609.23394](https://arxiv.org/abs/2609.23394).

G. Agrahari, C. Bibby, S. Boros, H. J. Garcia, F. Heidercheidt, Z. Wang. *On some structural properties of graphs with non-negative resistance curvature*. (2026). [arXiv:2607.13169](https://arxiv.org/abs/2607.13169).

M. Fairchild, H. J. Garcia, J. Murphy, H. Percle. *On the non-orientable 4-genus of torus knots with braid index 5 or 6*. *Journal of Knot Theory and Its Ramifications* **35** (2026), No. 7, 2650001. [doi:10.1142/S021821652650001X](https://doi.org/10.1142/S021821652650001X).

Talks

On some structural properties of graphs with non-negative resistance curvature Fall 2026

LSU Combinatorics Seminar

Noncontractive Function Systems Spring 2023

6th TX-LA Mathematics Undergraduate Conference

On The “Kelly Sequence” Spring 2023

6th TX-LA Mathematics Undergraduate Conference

Teaching

MATH 1431, Business Calculus — Lab Leader/TA, LSU Spring 2026
Sections 7, 11, 19

MATH 1431, Business Calculus — Lab Leader/TA, LSU Fall 2025
Sections 7, 8, 9, 11, 25, 28

Honors Calculus 1 — Primary Instructor, Arden Cahill Academy Fall 2023–Spring 2024

Mentoring

Mentor, [Directed Reading Program](#) Fall 2026

TA, BSM REU at Haverford College Summer 2026

TA, 2026 LSU Undergraduate FRC	Summer 2026
Mentor, Vertically Integrated Research Courses, LSU	Fall 2025–Spring 2026

Organisations

Organiser, LSU Informal Combinatorics Seminar	Fall 2025–present
Webmaster, LSU AMS Student Chapter	Fall 2026
Webmaster, LSU RTG	Spring 2026–present
Manager, LSU Math 3D Print Lab	Spring 2026–present
Officer, LSU AWM Student Chapter <i>President (Fall 2025–Spring 2026); Secretary (Spring 2026–present);</i> <i>Webmaster (Fall 2025–present)</i>	Fall 2025–present

Fellowships and Awards

LSU RTG Fellow	Fall 2026–present
Pasquale Porcelli Graduate Student Academic Excellence Award	Spring 2026
Robert and Betti Giles Senior Mathematics Award	Spring 2025
Peg and Tom Madden Mathematics Research Scholarship	Spring 2024
Porcelli Junior Scholarship	Spring 2023